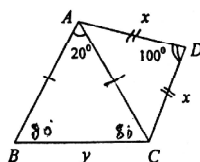


AMTI (NMTC) - 2004

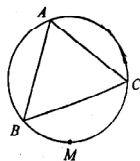
RAMANUJAN CONTEST - INTER LEVEL

1. When $1^{2003} + 2^{2003} + 3^{2003} + \dots + 2003^{2003}$ is divided by 2004, then the remainder is
 (A) 0 (B) 1 (C) 1002 (D) 2003
2. Each number from 1, 2, 3, ..., 100 (decimal scale) is written in base 6 and their product is also written in base 6. Then the number of zeroes at the end of this product is
 (A) 24 (B) 48 (C) 18 (D) 97
3. Given that $a_1, a_2, \dots, a_{2004}$ are distinct positive real numbers then $\frac{a_1}{a_2} + \frac{a_2}{a_3} + \dots + \frac{a_{2003}}{a_{2004}} + \frac{a_{2004}}{a_1}$ is
 (A) less than 2004 (B) less than 1
 (C) greater than 2004 (D) equal to 1
4. Two chords of a circle bisect each other. Then
 (A) The chords are of equal length but less than the length of the diameter.
 (B) The chords are of unequal length.
 (C) The chords should be perpendicular bisectors of each other.
 (D) The chords are any two diameters.
5. A man tosses a fair coin till he gets a head. The probability that he gets head in at most n tosses is
 (A) $\frac{1}{n}$ (B) $\frac{1}{2^n}$ (C) $1 - \frac{1}{n}$ (D) $1 - \frac{1}{2^n}$
6. ----
 (A) --- (B) --- (C) --- (D) ---
7. ----
 (A) --- (B) --- (C) --- (D) ---
8. ----
 (A) --- (B) --- (C) --- (D) ---
9. ----
 (A) --- (B) --- (C) --- (D) ---
10. ----
 (A) --- (B) --- (C) --- (D) ---
11. Kumar never lies except on Tuesdays. On Tuesdays, he always lies. On how many days of the week can he say "If I did not lie yesterday then I will lie tomorrow."
 (A) 1 (B) 2 (C) 3 (D) 4
12. The range of function $f(r) = {}_{7-r}P_{r-3}$ where ${}_mP_n$ represents permutations of 'm' things taken 'n' at a time and where r is a non-negative integer is
 (A) {1, 2, 3, 4, 5} (B) {1, 2, 3, 4} (C) {1, 2} (D) {1, 2, 3}
13. The roots of $64x^3 - 144x^2 + 92x - 15 = 0$ are in A.P. Then the difference between the largest and smallest is
 (A) $\frac{1}{2}$ (B) $\frac{3}{4}$ (C) $\frac{7}{8}$ (D) 1
14. In the following figure, if $AD = DC = x$ and $BC = y$ then $AB = AC = ?$



- (A) $x + y$ (B) $2y + x$ (C) $2y$ (D) $2x$

15. ABCD is a tetrahedron. The number of planes from which the distances to A, B, C, D are equal is
 (A) 0 (B) 5 (C) 4 (D) 3
16. The three last digits of 7^{999} are
 (A) 263 (B) 143 (C) 343 (D) 523
17. Each of the faces of a cube is coloured by a different colour. How many of the colourings are distinct ?
 (A) 6 (B) 30 (C) 18 (D) 24
18. Which of the following is a continuous function f satisfying $3f(2x + 1) = f(x) + 5x$?
 (A) $f(x) = 2x + 5$ (B) $f(x) = x + 1$ (C) $f(x) = x - \frac{3}{2}$ (D) $f(x) = x + \frac{1}{2}$
19. A, B, C, M are four points on a circle such that ABC forms an equilateral triangle and M lies on the arc BC as in the figure. Which of the following holds good ?



- (A) $|MA| + |MB| + |MC| = 2|AB|$ (B) $|MA| = |MB| + |MC|$
 (C) $|MA|^2 = |MB|^2 + |MC|^2$ (D) $|MA|^2 + |MB|^2 + |MC|^2 = \text{Area of } \triangle ABC$
20. The positive numbers x and y satisfy $xy = 1$. The minimum value of $\frac{1}{x^4} + \frac{1}{4y^4}$ is
 (A) $\frac{1}{2}$ (B) $\frac{5}{8}$ (C) 1 (D) $\frac{5}{4}$
21. A cubic polynomial P is such that $P(1) = 1$, $P(2) = 2$, $P(3) = 3$ and $P(4) = 5$. The value of $P(6)$ is
 (A) 7 (B) 10 (C) 13 (D) 16
22. In $\triangle ABC$, the altitude from A to BC meets BC at D and the altitude from B to CA meets AD at H. If $AD = 4$, $BD = 3$, $CD = 2$ then the length of HD is
 (A) $\frac{\sqrt{5}}{2}$ (B) $\frac{3}{2}$ (C) $\sqrt{5}$ (D) $\frac{5}{2}$
23. A fair coin is tossed 10,000 times. The probability p of obtaining at least 3 heads in a row satisfies
 (A) $0 \leq p \leq \frac{1}{4}$ (B) $\frac{1}{4} \leq p \leq \frac{1}{2}$ (C) $\frac{1}{2} \leq p < \frac{3}{4}$ (D) $\frac{3}{4} \leq p < 1$
24. The number of different positive integer triplets (x, y, z) satisfying the equations $x^2 + y - z = 100$ and $x + y^2 - x = 124$ is
 (A) 0 (B) 1 (C) 2 (D) 3
25. If $\phi(n)$ denotes the number of positive integers less than n and prime to n and $A = \{n | \phi(n) = 17\}$ then the number of elements in A is
 (A) infinite (B) 16 (C) 17^2 (D) zero
26. For what values of n is an n -digit number uniquely determined from the sum and product of its digits ?
 (A) 1 (B) 2 (C) 3 (D) 4
27. The number of polynomials $P(x)$ satisfying the equation $P(x^2) + 2x^2 + 10x = 2xP(x + 1) + 3$ is
 (A) 2 (B) 1 (C) 3 (D) infinite
28. Let n be a positive integer. Then the number of common factors of $n^2 + 3n + 1$ and $n^2 + 4n + 3$ is
 (A) $(n + 1)$ (B) $n - 1$ (C) 2 (D) 1

29. If $f(n + 1) = f(n) + n$ for all $n \geq 0$ or $f(0) = 1$ then $f(200)$ equals
(A) 21100 (B) 21000 (C) 20900 (D) 19900
30. The rays PX and PY cut off arc AB and CD of a circle with radius 4. If the length of arc CD is 2 times the length of the arc AB and length of CD is $\frac{4\pi}{5}$, then the angle APB is
(A) 9° (B) 10° (C) 12° (D) 18°
